MSG No. 72.544 $Iba'm'$ [Type III, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `22'2'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,5,6]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[3,4,7,8]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[9,10,13,14]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[11,12,15,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `22'2'`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[1,2,5,6]`
2	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[3,4,7,8]`
3	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[9,10,13,14]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[11,12,15,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `..2'/m'`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,3,6,8]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[2,4,5,7]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,11,14,16]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[10,12,13,15]`

Table 4: Wyckoff site: `4d`, site symmetry: `..2'/m'`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,3,6,8]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[2,4,5,7]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,11,14,16]`
4	$\left[0,\ \frac{1}{2},\ 0\right]$	`[10,12,13,15]`

Table 5: Wyckoff site: `8e`, site symmetry: `-1`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,11]`
2	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[2,12]`
3	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[3,9]`
4	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[4,10]`
5	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[5,15]`
6	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[6,16]`
7	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[7,13]`

8	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[8,14]`

Table 6: Wyckoff site: `8f`, site symmetry: `2..`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,2]`
2	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[3,4]`
3	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[5,6]`
4	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[7,8]`
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[9,10]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[11,12]`
7	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[13,14]`
8	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[15,16]`

Table 7: Wyckoff site: `8g`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{4}\right]$	`[1,5]`
2	$\left[0,\ -y,\ \frac{1}{4}\right]$	`[2,6]`
3	$\left[0,\ -y,\ \frac{3}{4}\right]$	`[3,7]`
4	$\left[0,\ y,\ \frac{3}{4}\right]$	`[4,8]`
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{3}{4}\right]$	`[9,13]`
6	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[10,14]`
7	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	`[11,15]`
8	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[12,16]`

Table 8: Wyckoff site: `8h`, site symmetry: `..2'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,6]`
2	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[2,5]`
3	$[0,\ 0,\ -z]$	`[3,8]`
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[4,7]`
5	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,14]`
6	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[10,13]`
7	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[11,16]`
8	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[12,15]`

Table 9: Wyckoff site: `8i`, site symmetry: `..2'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,6]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[2,5]`
3	$\left[0,\ \frac{1}{2},\ -z\right]$	`[3,8]`

continued ...

Table 9

No.	position	mapping
4	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[4,7]
5	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	[9,14]
6	$\left[\frac{1}{2},\ 0,\ -z\right]$	[10,13]
7	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	[11,16]
8	$\left[\frac{1}{2},\ 0,\ z\right]$	[12,15]

Table 10: Wyckoff site: 8j, site symmetry: ..m'

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,8]
2	$\left[x,\ -y,\ \frac{1}{2}\right]$	[2,7]
3	$[-x,\ -y,\ 0]$	[3,6]
4	$\left[-x,\ y,\ \frac{1}{2}\right]$	[4,5]
5	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[9,16]
6	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	[10,15]
7	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[11,14]
8	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ 0\right]$	[12,13]

Table 11: Wyckoff site: 16k, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	[2]
3	$[-x,\ -y,\ -z]$	[3]
4	$\left[-x,\ y,\ z+\frac{1}{2}\right]$	[4]
5	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	[5]
6	$[-x,\ -y,\ z]$	[6]
7	$\left[x,\ -y,\ z+\frac{1}{2}\right]$	[7]
8	$[x,\ y,\ -z]$	[8]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[9]
10	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[10]
11	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[11]
12	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	[12]
13	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$	[13]
14	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[14]
15	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	[15]
16	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[16]